Project Report

on

Health Analysis using Microsoft Power Bi and Python

Submitted in partial fulfillment of the requirement for the award of degree of

Master of Computer Application (MCA)

at

University of Mumbai

Submitted by

Mr. PATIL PRATHMESH SURESH

Under the guidance of

prof. Nidhi Poonia

Bharati Vidyapeeth's

Institute of Management and Information Technology

Sector 8, CBD Belapur

Navi Mumbai

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Bharati Vidyapeeth's

Institute of Management and Information Technology

Navi Mumbai

Certificate of Approval

This is to certify that the Project titled **' Health Analysis Using Microsoft Power Bi and Python '** is successfully done by **Mr. Prathmesh Patil** during internship of his course in partial fulfillment of **Master's of Computer Application** under the **University of Mumbai, Mumbai**, through the Bharati Vidypeeth's Institute of Management and Information Technology, Navi Mumbai carried out by him under our guidance and supervision.

Sign & Date
Guide

External Examiner
Signature and Date

Principal
Dr.Suhasini Vijaykumar

College Seal

Declaration

I the undersigned solemnly declare that the project report is based on my own work carried out during the MCA course under the Guidance of Mrs. Nidhi Poonia.

The work has not been submitted partly or wholly to any other Institution for any other degree/diploma in this university or any other University.

We have followed the guidelines provided by the university in writing the report. Whenever we have used materials (data, theoretical analysis, text, figures and images) from other sources, we have given due credit to them in the text of the report and giving their details in the references.

(Signature)

(Mr. Prathmesh Patil)

Date: 30th June 2023

Acknowledgement

I avail this opportunity to express my sincere and deep gratitude to many who are a factor in helping me gain the knowledge and experience during the project and throughout the course. I have great pleasure in presenting this project. The completion of this project is not merely due to only my own efforts but also due to the guidance given by our professors. I am thankful to our project guide Prof. Nidhi Poonia for her support. I also thank Dr. Suhaisni Vijaykumar, Incharge Principal BVIMIT and respected faculty members for their kind support and help throughout the entire course.

Finally I express my deep regards to all of those who stretch their helping hand in the execution of my project.

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Date: 30th June 2023

Mr. Prathmesh Patil

Abstract

Analysis on the data collected through Google form based on the health problems of people. Analysis would be done according to particular region and age groups. The overall health of a person would be analyzed or visualized using Microsoft Power Bi.

Health Data Analysts take the health data being collected and use skills such as data acquisition, data management, data analysis and data interpretation to provide actionable insights. The role of big data in health care management and the increasing need for improvement in the health care sector has led to a higher demand for qualified health data analysts.

The role of a health data analyst varies based on their position and industry of choice. Regardless of industry, a health data analyst will need to be able to do the following tasks.

o Collect or mine data

o Examine current and historical data

o Evaluate raw data

o Build predictive models

o Automate reports

Thus, we will try to achieve these mentioned goals.

Contents

Chapter 1

Introduction

1.1 Introduction of the project

Data analytics is the process of examining raw datasets to find trends, draw conclusions and identify the potential for improvement. Health care analytics uses current and historical data to gain insights, macro and micro, and support decision-making at both the patient and business level. The use of health data analytics allows for improvements to patient care, faster and more accurate diagnoses, preventive measures, more personalized treatment and more informed decision-making. Health data is any data relating to the health of an individual patient or collective population. This information is gathered from a series of health information systems (HIS) and other technological tools utilized by health care professionals, insurance companies and government organizations. We are able to see a holistic view of each individual patient as well as trends tied to location, socioeconomic status, race and predisposition. The information being collected can be broken down into specific datasets that can then be analyzed. With digital data collection, there is more and more health care data to be analyzed every second. With the increase of electronic record keeping, applications and other electronic means of data collection and storage, there is a significant amount of data being collected in real time.

The World Health Organization (WHO) declared COVID-19 a global pandemic in March 2020; there have been in excess of 92 million confirmed cases worldwide. International responses were implemented to contain the spread of the virus, with various states of imposed lockdown applied in most countries. There are many psychosocial consequences of pandemics, with research showing that individuals' mental health has been severely affected by COVID-19 and the associated lockdown. Some groups are particularly vulnerable including those who contract the disease, those at a heightened risk of contraction and people with pre-existing medical, psychiatric or substance use problems. Deteriorating mental health also has an adverse affect on individual's physical health. By sitting at home, people were deprived of outdoor activities. Many unhealthy habits were inculcated amongst the people. Thus, this leads to collect data of people and analyze their general health.

In the previous semester the aim of the project was to do analysis on the data collected through Google form based on the health problems of people. Analysis was done according to particular region and diseases. Visualization of the data was done using Microsoft Power Bi tool. In this project with the help of machine learning algorithm we will predict an outcome from the collected data. Machine Learning algorithms are the programs that can learn the hidden patterns from the data, predict the output, and improve the performance from experiences on their own. Different algorithms can be used in machine learning for different tasks, such as simple linear regression that can be used for prediction problems like stock market prediction, and the KNN algorithm can be used for classification problems. Machine learning algorithms are of the following types:

1. Supervised Learning Algorithms: Supervised learning is a type of Machine learning in which the machine needs external supervision to learn. The supervised learning models are trained using the labeled dataset. Once the training and processing are done, the model is tested by providing a sample test data to check whether it predicts the correct output. The goal of supervised learning is to map input data with the output data.

2. Unsupervised Learning Algorithms: It is a type of machine learning in which the machine does not need any external supervision to learn from the data, hence called unsupervised learning. The unsupervised models can be trained using the unlabeled datasets that is not classified, nor categorized, and the algorithm needs to act on that data without any supervision. In unsupervised learning, the model doesn't have a predefined output, and it tries to find useful insights from the huge amount of data. These are used to solve the Association and Clustering problems.

1.2 Objectives of the Study

The objective of the project is to use a machine learning algorithm over the collected data and predict the outcome. A decision tree would be shown as an output of the processed data. The processing would be done on the collected attributes such as region, age groups and overall health habits. This kind of processing is also called as prescriptive analysis.

Prescriptive Analytics:
Prescriptive analytics will also make predictions about future outcomes. Machine learning is a big factor with this type of analytics. The information provided can help determine the best course of action. We can gain insight on what course of action should be taken to reach the most ideal outcome with prescriptive analytics.

In this project we are going to use the CART algorithm for the prediction. CART is a predictive algorithm used in Machine learning and it explains how the target variable's value can be predicted based on other matters. It is a decision tree where each fork is split into a predictor variable and each node has a prediction for the target variable at the end. In the decision tree, nodes are split into sub-nodes on the basis of a threshold value of an attribute. The root node is taken as the training set and is split into two by considering the best attribute and threshold value. Further, the subsets are also split using the same logic. This continues till the last pure sub-set is found in the tree or the maximum number of leaves possible in that growing tree.

Advantages of CART:
- Results are simplistic.
- Classification and regression trees are Nonparametric and Nonlinear.
- Classification and regression trees implicitly perform feature selection.
- Outliers have no meaningful effect on CART.
- It requires minimal supervision and produces easy-to-understand models.

1.3 Scope of the Project

Health data is any data relating to the health of an individual patient or collective population. We are able to see a holistic view of each individual patient as well as trends tied to location, socio-economic status, race and predisposition. The information being collected can be broken down into specific datasets that can be then analyzed. Big data and health data analytics have played an integral role in fight against Covid-19. Analyzing the health data has allowed for a better understanding of how to respond and treat patients. Hence, by analyzing the data we will be able to see which age group has unhealthy lifestyle.

This visualization will help in understanding the overall health of an individual hence one would be able to get timely treatment and can help people understand more about healthy lifestyle.

The algorithm used in the project will predict whether the individual is in good physical health based on the attributes collected.

Chapter 2

Review of Literature Survey

There are several reasons why conducting a health analysis project can be beneficial, including:

Identifying health trends: A health analysis project can help identify health trends in a particular population, such as the prevalence of certain diseases, risk factors for certain conditions, and patterns in health behaviors.

Informing public health policies: The information gathered from a health analysis project can help inform public health policies and interventions aimed at improving health outcomes in a particular community or population.

Evaluating interventions: Health analysis projects can help evaluate the effectiveness of interventions aimed at improving health outcomes, such as health education programs, screening programs, and treatment programs.

Identifying health disparities: A health analysis project can help identify disparities in health outcomes across different populations, such as differences in access to healthcare, socioeconomic status, or race/ethnicity.

Planning for future healthcare needs: The information gathered from a health analysis project can help healthcare providers and policymakers plan for future healthcare needs, such as identifying areas where additional healthcare resources are needed or forecasting future healthcare demands.

Overall, a health analysis project can provide valuable insights into the health status of a population and inform strategies for improving health outcomes.

2.1 The current State of the System in the Market

There are various current systems for health analysis, including:

Electronic Health Records (EHRs): EHRs are digital records of a patient's health information, including medical history, laboratory results, and medication prescriptions. EHRs allow for the collection, storage, and analysis of vast amounts of health data.

Health Information Exchanges (HIEs): HIEs are networks that enable the sharing of health information between healthcare providers, such as hospitals, clinics, and pharmacies. HIEs facilitate the coordination of care and can improve the accuracy and completeness of health data.

Clinical Decision Support Systems (CDSS): CDSSs are computer programs that provide healthcare providers with patient-specific information and knowledge to assist in clinical decision-making. CDSSs can suggest diagnoses, treatment plans, and medication options based on a patient's health history and current condition.

Population Health Management (PHM) Systems: PHM systems use data analysis tools to identify patterns and trends in population health. PHM systems can help healthcare providers better understand the health needs of their patient populations and develop targeted interventions to improve health outcomes.

Telehealth: Telehealth refers to the use of technology, such as video conferencing and remote monitoring devices, to provide healthcare services remotely. Telehealth can improve access to care, reduce healthcare costs, and increase patient convenience.

Health Analytics: Health analytics involves the analysis of large datasets to identify patterns and insights related to healthcare delivery and outcomes. Health analytics can help healthcare providers optimize care delivery and improve health outcomes for patients.

Chapter 3

Domain Knowledge

• Domain overview:
Data visualization and analysis is a domain that involves the exploration, interpretation, and communication of data through visual representations. It is a field that combines statistical analysis, computer science, and design to create meaningful and informative visualizations that allow data analysts and decision makers to identify patterns, relationships, and insights within large and complex data sets.

Data visualization and analysis tools and techniques enable data analysts to process and extract valuable insights from large amounts of data more efficiently and effectively than traditional methods. These insights can be used to support evidence-based decision making in a wide range of fields, including business, healthcare, finance, social sciences, and many others.

Some popular data visualization and analysis tools and techniques include charts, graphs, maps, dashboards, and infographics. These tools help to convey complex information in a clear and understandable way, making it easier for decision makers to understand and act on the insights gained from data analysis.

Data analysis is the process of inspecting, cleaning, transforming, and modeling data to uncover useful information, draw conclusions, and support decision-making. It involves applying statistical and mathematical techniques to extract insights from raw data, identify patterns and relationships, and generate visualizations and reports that communicate findings in a clear and actionable way. Data analysis is used in various fields such as business, science, healthcare, and social sciences to make informed decisions, improve processes, and optimize performance.

Machine learning is a subfield of artificial intelligence (AI) that focuses on the development of algorithms and models that enable computers to learn from and make predictions or decisions based on data, without being explicitly programmed for each specific task. The fundamental idea behind machine learning is to create algorithms that automatically improve their performance over time by learning from experience.

In traditional programming, humans write explicit instructions for computers to follow. However, in machine learning, computers learn from data patterns and examples to identify relationships, make predictions, or solve complex problems. The learning process involves training a model on a dataset, which typically consists of input data (features) and corresponding output data (labels or target variables). The model then learns patterns and generalizes from the training data to make predictions or decisions on new, unseen data.

There are several types of machine learning algorithms, including:
1. Supervised Learning: In this approach, the algorithm learns from labeled training data, where each input example is associated with a corresponding correct output. The algorithm aims to learn a mapping function that can predict outputs for new inputs accurately.

2. Unsupervised Learning: Here, the algorithm learns from unlabeled data, finding patterns or structures within the data without specific guidance. Clustering and dimensionality reduction are common tasks in unsupervised learning.

3. Reinforcement Learning: This type of learning involves an agent interacting with an environment and learning to make decisions or take actions to maximize a reward signal. The agent receives feedback in the form of rewards or punishments, enabling it to learn from its actions and improve its decision-making abilities over time.

Machine learning has found applications in various domains, including image and speech recognition, natural language processing, recommendation systems, fraud detection, autonomous vehicles, healthcare, finance, and many others. It is a powerful tool for extracting insights, making predictions, and automating tasks based on patterns and data-driven reasoning.

The CART algorithm, which stands for Classification and Regression Trees, is a machine learning algorithm used for both classification and regression tasks. It is a decision tree-based algorithm that recursively partitions the input data based on feature values to create a tree-like model for making predictions.

The CART algorithm builds a binary tree structure, where each internal node represents a decision based on a specific feature, and each leaf node represents a predicted class label or a numerical value. The tree is constructed by selecting the best feature and a corresponding splitting point that optimally separates the data based on certain criteria.

For classification tasks, the CART algorithm aims to minimize impurity or maximize information gain. Common impurity measures include Gini impurity and entropy. The algorithm evaluates different feature splits and selects the one that results in the greatest reduction in impurity or the highest information gain.

For regression tasks, the CART algorithm typically minimizes the mean squared error (MSE) or a similar metric to find the best feature and splitting point. The goal is to create partitions that minimize the variance within each partition. Once the tree is built, making predictions involves traversing the tree from the root node to a leaf node based on the feature values of the input data. The predicted class label or numerical value associated with the leaf node reached becomes the final prediction.

CART trees have desirable properties, such as interpretability, as the decision-making process can be easily visualized and understood. Additionally, they can handle both numerical and categorical features and are relatively robust to outliers and missing data. Moreover, CART trees can be extended to ensemble methods like random forests and gradient boosting, where multiple trees are combined to improve predictive performance.

Overall, the CART algorithm is a flexible and widely used approach in machine learning for decision-making tasks, offering both interpretability and good predictive capabilities.

Chapter 4

Technological Development

4.1 Software Requirements

These are the software requirements for the project:
Tool used: Jupyter Notebook, Visual Studio Code, Microsoft Power BI
Operating system: Windows 11/10/8/7 64-bit operating system

4.2 Hardware Requirements

These are the minimum hardware requirements for the project: Ram: 4 GB or higher.
Processor: 9th Gen Intel(R) Core(TM) i5-12400 2.50 GHz or higher.
HDD: 256 GB or higher.
Monitor: 1024 x 768 minimum screen resolution.

Chapter 5

System Study

5.1 Advantages of Data Analysis

Data visualization communicates information faster than traditional reports. A more concise, more refined language, visualized data can be easily sorted to show a quick overview of whatever information is needed. The importance of data visualization in sales is also important. Charts, graphs, and visual representations can be processed faster, support visual learners, and show insights and actionable items that lead to increased sales productivity.

1. Visualized Data Is Processed Faster:

Visual content is processed much faster and easier than text. Data visualization taps into this concept of how quickly our brains can recognize images and make sense of them. data visualization tools play into our biological sweet spot. The human mind may not intuitively understand complex statistical models or things like 'R squared' values, but we are quite adept at picking out patterns from visual displays."

2. Data Visualization Dashboards Support Visual Learners:

While 90 percent of information submitted to the brain is visual, learning styles vary among the population. Data visualization and online data visualization tools help make it possible to quickly comprehend the information presented. Moving past the spreadsheet era, modern technology has transformed information from generic spreadsheets into appealing and easy-to-read charts and graphs. online data visualization is a tool to present data visually and gain insights from that data.

3. Data visualization tools show insights that may be missed in traditional reports :
Getting the entire company in the habit of seeing the dashboard reports and data visualization can help get a better picture of the organization. Once your organization identifies the information and key performance indicators (KPIs) they'd like to have visualized, CRM can produce the visualizations. Examples of data visualization reports might include: Sales by period, sorted by sales person or product, deals in the pipeline, sorted by individual accounts; call time per customer; or deals closed by sales rep.

4. Data visualization gives actionable items:
Data visualization may help your organization see where there's room for improvement or where performance is high. Actionable items can result by identifying successes and areas for improvement. For example, if your sales team knows that for every X number of calls, Y number of sales will result, creating a visual report based on calls per sales rep and progress to call goal is a visual motivator to meet the call quota. Similarly, a pipeline report showing where each deal falls along the sales pipeline shows sales teams the next steps to be taken.

5. Data visualization increases productivity and sales :
Being able to visualize data produces real results. The time saved in creating up-to-date reports means greater efficiency company-wide. The study also reports that 48 percent of business intelligence users at companies with visual data discovery are able to find information they need without the help of IT staff all or most of the time.

5.2 Disadvantages of Data Analysis

Lack of alignment within teams:

There is a lack of alignment between different teams or departments within an organization. Data analytics may be done by a select set of team members and the analysis done may be shared with a limited set of executives. However, the insights generated by these teams are either of not much value or are having limited impact on organizational metrics. This could be due to a "silos" way of working with each team only using their existing processes disconnected from other departments. The analytics team should be focused on answering the right questions for the business and the results generated by data analytics teams needs to be properly communicated to the right employees to drive the right set of actions and behaivour so that it can have an positive impact on the organization.

5.3 Advantages of Machine Learning

Machine learning offers several advantages that contribute to its popularity and broad range of applications. Here are some key advantages of machine learning:

Automation and Efficiency: Machine learning enables automation of tasks that would typically require manual effort or complex rule-based programming. By learning from data patterns and examples, machines can perform tasks faster, more accurately, and at scale, leading to increased efficiency and productivity.

Handling Complex and Large-scale Data: With the exponential growth of data in various forms, such as text, images, and sensor readings, machine learning provides techniques to process, analyze, and extract valuable insights from large and complex datasets. It can identify patterns, correlations, and hidden structures that may be challenging or impossible for humans to discern manually.

Improved Decision Making: Machine learning algorithms can analyze vast amounts of data and learn complex relationships to make informed predictions or decisions. By considering a broader range of factors and patterns, machine learning models can often provide more accurate and consistent decision-making support than traditional methods.

Adaptability and Learning: Machine learning models have the ability to adapt and improve their performance over time as they are exposed to more data. They can continuously learn from new examples, update their knowledge, and adjust their predictions accordingly. This adaptability makes machine learning models suitable for dynamic environments where patterns may change or evolve.

Discovery of Insights and Patterns: Machine learning techniques can uncover hidden insights, trends, and patterns in data that may not be apparent through traditional analysis methods. These discoveries can lead to valuable knowledge and actionable insights, enabling organizations to gain a competitive edge, make data-driven decisions, and discover new opportunities.

Personalization and Recommendation Systems: Machine learning powers personalized recommendations in various domains, such as e-commerce, streaming services, and content platforms. By analyzing user preferences, behavior, and historical data, machine learning algorithms can tailor recommendations and suggestions to individual users, enhancing user experiences and engagement.

Automation of Repetitive Tasks: Machine learning can automate repetitive and mundane tasks, freeing up human resources to focus on more complex and creative endeavors. This automation can lead to increased productivity, reduced errors, and improved job satisfaction.

Wide Range of Applications: Machine learning finds applications in numerous domains, including healthcare, finance, marketing, cybersecurity, robotics, natural language processing, image recognition, and more. Its versatility allows for solving diverse problems and driving innovation across various industries.

It's important to note that while machine learning offers numerous advantages, careful consideration must be given to data quality, bias, interpretability, and ethical considerations to ensure responsible and meaningful use of machine learning techniques.

5.4 Features of current system

Microsoft Power BI is a business analytics service that provides a suite of tools for data visualization, data analysis, and data sharing. Some of its key features include:

1. Data connectivity: Power BI can connect to a wide range of data sources, including Excel files, cloud-based and on-premises databases, and popular third-party applications such as Sales force, Google Analytics, and SharePoint.

2. Data transformation and modeling: Power BI allows users to transform and

shape data from multiple sources, create calculated columns and measures, and create relationships between tables to create data models.

3. Data visualization: Power BI provides a wide range of customizable data visualizations, including charts, graphs, maps, and tables. Users can also create interactive dashboards and reports to share with others.

4. Collaboration and sharing: Power BI allows users to collaborate on reports and dashboards, share them securely with others, and embed them in websites and other applications.

5. Natural language processing: Power BI provides natural language processing capabilities that allow users to ask questions about their data in plain English and receive visualizations and insights in response.

One of the key features of Power bi and the one that got its popularity is its wide range of visualizations. In Power bi, you can make visualizations as basic as a:

- Bar chart
- Pie chart

and as advanced as a:

- Histogram
- Gantt chart
- Bullet chart
- Motion chart
- Tree map
- Boxplot

and many more.

The algorithm was implemented using Jupyter Notebook.
Jupyter Notebook is an open-source web application that allows you to create and share documents that contain live code, equations, visualizations, and explanatory text. Here are some key features of Jupyter Notebook:

Interactive Environment: Jupyter Notebook provides an interactive environment where you can write and execute code in individual cells. This allows for easy experimentation, iterative development, and immediate feedback.

Support for Multiple Programming Languages: Jupyter Notebook supports a wide range of programming languages, including Python (the most commonly used), R, Julia, and more. This makes it versatile and accessible for data analysis, machine learning, scientific computing, and other domains.

Code Execution: You can execute code cells individually or run the entire notebook. This flexibility allows for step-by-step execution, debugging, and testing of code snippets, which promotes an interactive and exploratory coding experience.

Inline Plots and Visualizations: Jupyter Notebook enables the generation of plots, charts, and visualizations directly within the notebook. By using libraries such as Matplotlib, Seaborn, or Plotly, you can visualize data, explore patterns, and communicate insights effectively.

Collaboration and Sharing: Jupyter Notebook files (.ipynb) can be shared and collaborated on with others. It allows you to export notebooks in various formats, including HTML, PDF, and Markdown, making it easy to share and present your work.

Integration with Data Science Libraries: Jupyter Notebook integrates seamlessly with popular data science libraries and frameworks such as NumPy, Pandas, SciPy, scikit-learn, TensorFlow, and PyTorch. This makes it a powerful tool for data manipulation, analysis, and machine learning workflows.

These features make Jupyter Notebook a popular choice among data scientists, researchers, and educators for interactive coding, data exploration, and sharing computational workflows.

Chapter 6

Problem Definition

6.1 Problem definition of project

In this semester the aim of the project is to use a machine learning algorithm over the collected data and predict the outcome. The outcome can be predicted over the attributes collected such as region, age groups and overall healthy habits of the people. The model will predict the physical health of an individual based on the attributes provided to the model.
The project has been implemented using CART algorithm for prediction. Health Data Analysts take the health data being collected and use skills such as data acquisition, data management, data analysis and data interpretation to provide actionable insights. The role of big data in health care management and the increasing need for improvement in the health care sector has led to a higher demand for qualified health data analysts. The role of a health data analyst varies based on their position and industry of choice. Regardless of industry, a health data analyst will need to be able to do the following tasks.

o Collect or mine data
o Examine current and historical data
o Evaluate raw data
o Build predictive models
o Automate reports
Thus, we will try to achieve these mentioned goals.

6.2 Use-Case Diagram

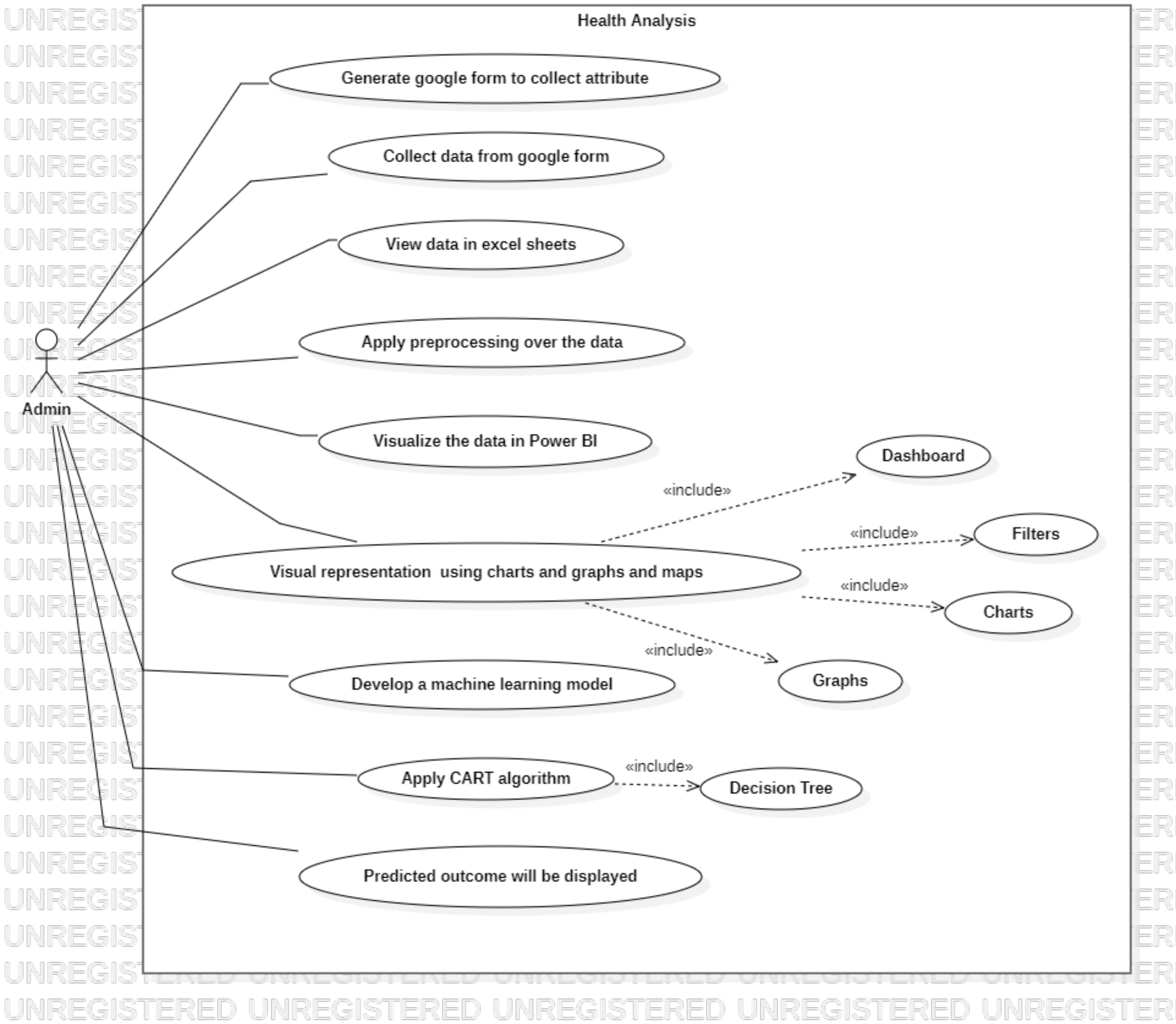


Figure 6.1: Use Case Diagram

Chapter 7

Requirement Gathering

There are several types to identify requirement , here we are going to consider functional requirements and the technical requirements for the proposed system.

Technical requirements
Functional requirements

Identify functional and technical requirements:

The functional requirement for the system are there should proper visualization of the project.
The visualization can be based on region or diseases. Efficient display of data is expected.

The technical requirements for the proposed system are of two types i.e.
Software requirements and Hardware requirements: The software requirements for the system are: Jupyter Notebook, Visual Studio Code, Microsoft Power BI and Operating system: Windows 11/10/8/7 64-bit operating system

The hardware requirements for the system are intel 3 processor, 4GB or more RAM , hardisk of 500GB or more and operating system of windows 7 and more.

7.1 Gantt Chart

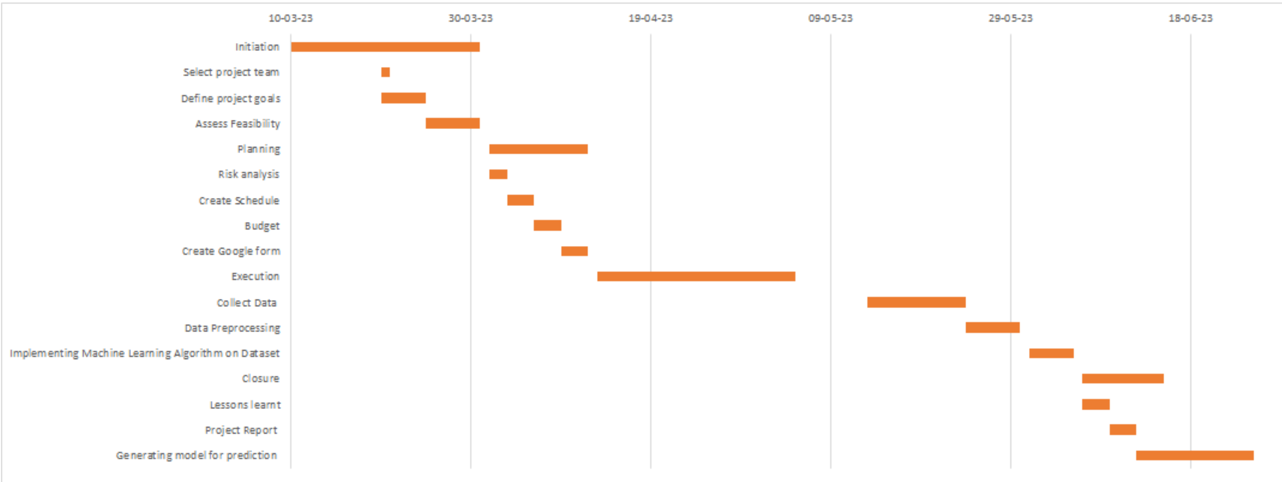


Figure 7.1: Gantt Chart

Chapter 8

Methodology

8.1 Waterfall Model

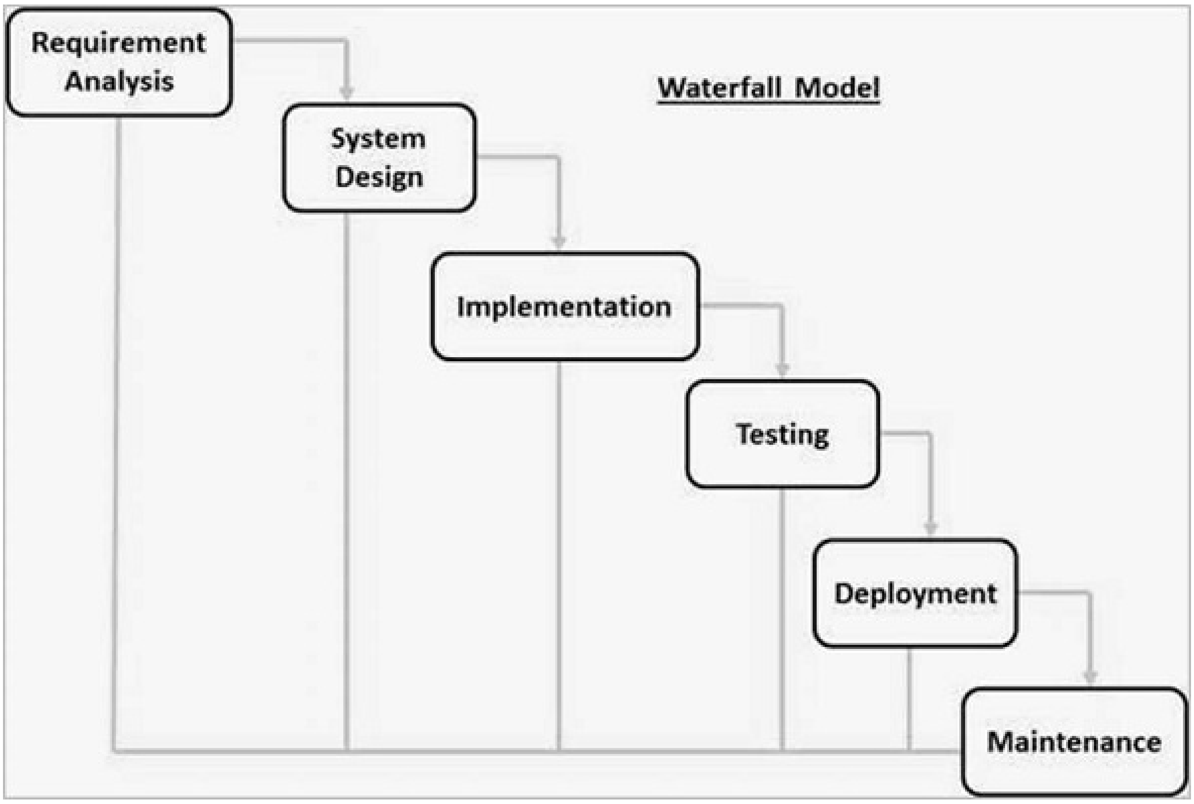


Figure 8.1: Waterfall Model

There is no specific SDLC model that is exclusively used for data analysis. The choice of SDLC model for data analysis depends on the specific project requirements, goals, and the development team's preferences.
The waterfall model is a sequential software development process, where the development process flows downwards, similar to a waterfall. Each phase of the process must be completed before the next one can begin. Here is an example of a waterfall model for developing a simple e-commerce website:

Requirements Gathering: The first phase is to gather requirements from the customer, such as what features the website should have, what products should be sold, and what payment options should be available.

Analysis: In this phase, the requirements are analyzed to ensure that they are clear, concise, and complete. The team identifies any potential challenges and determines how to address them.

Design: In this phase, the website design is created, including the layout, color scheme, and overall appearance. The design must align with the requirements gathered in the first phase. Implementation: This is where the development team builds the website, including coding, testing, and debugging. They must ensure that the website functions as intended and that it meets the design and requirements.

Testing: In this phase, the website is tested to ensure that it functions properly, that there are no bugs, and that it meets all of the requirements.

Deployment: Once the website is tested and approved, it is deployed to the production environment.

Maintenance: After deployment, the website must be maintained to ensure that it continues to function correctly and that any issues that arise are addressed promptly.

In summary, the waterfall model is a sequential software development process that moves from requirements gathering to maintenance. Each phase must be completed before the next one can begin, and the process is not cyclical. This model is useful for projects with well-defined requirements and a clear end goal, as it allows for a structured approach to development.

Chapter 9

Design Approach

9.1 Data Flow Diagram

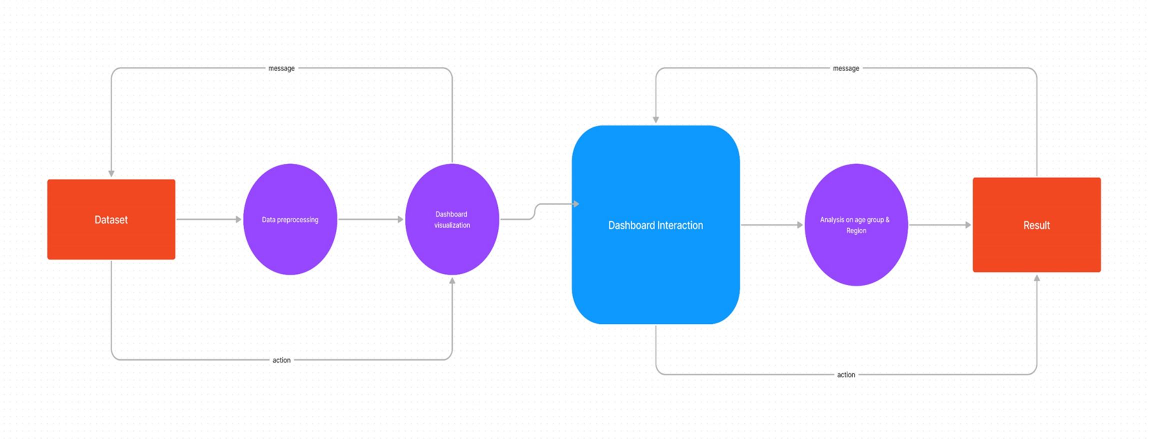


Figure 9.1: Data Flow Diagram

A data flow diagram (DFD) is a graphical representation of the flow of data within an organization or system. It is a modeling technique used to describe the flow of data and the processing of data within a system.

A DFD visually represents the system's inputs, outputs, and processes. The diagram typically consists of a series of interconnected boxes, known as processes, which represent the activities that transform the data. The data itself is represented by arrows, which indicate the flow of data from one process to another or from an external entity to a process.

The different levels of a DFD include context diagrams, which provide an overview of the system, and detailed diagrams that show the processes and data flows in more detail.

DFDs are commonly used in software engineering, systems analysis, and project management to help understand the flow of data through a system and to aid in the design of new systems or the modification of existing systems.

9.2 Implementaion of CART algorithm

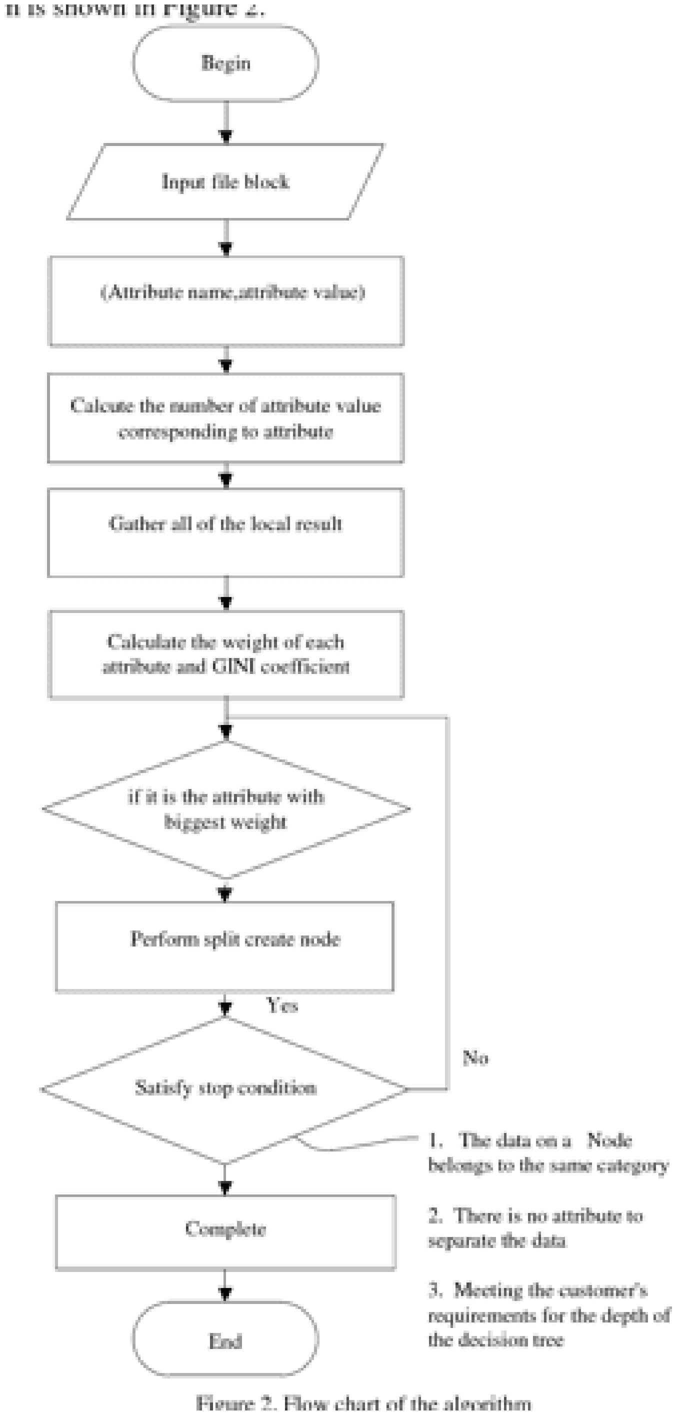
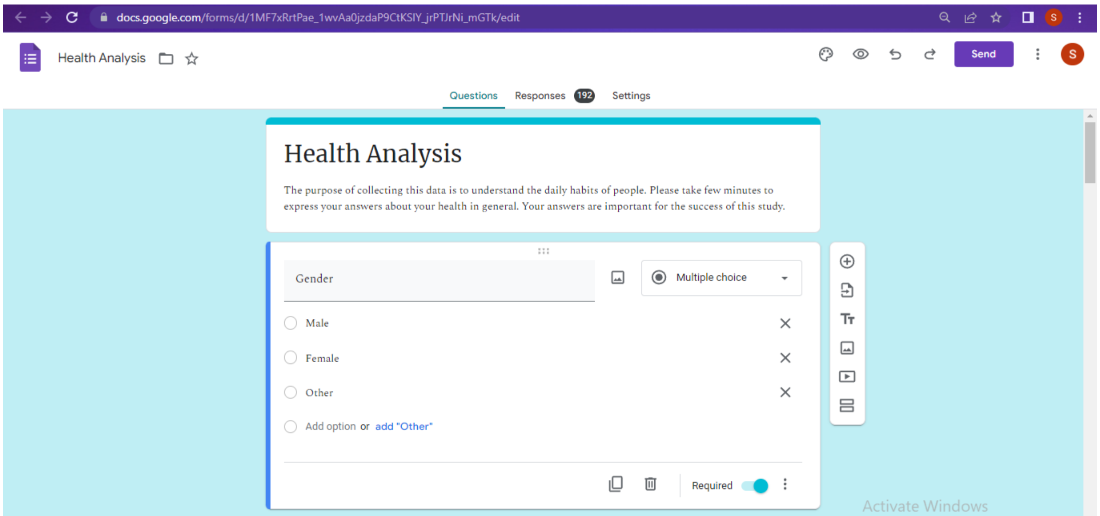


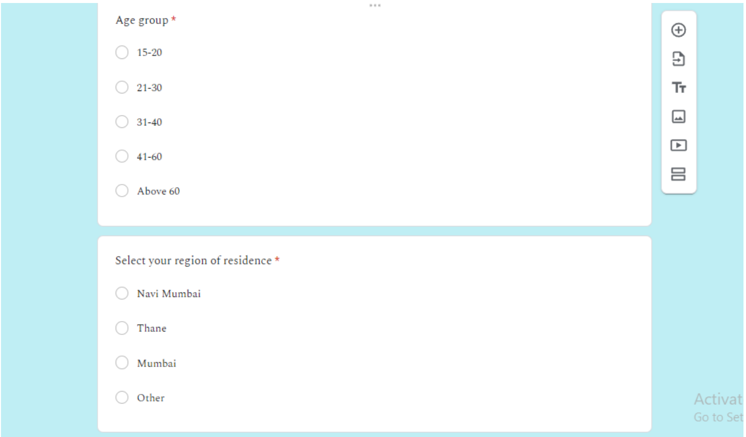
Figure 9.2: Flow Chart of CART algorithm

Chapter 10

Implementation

10.1 Form



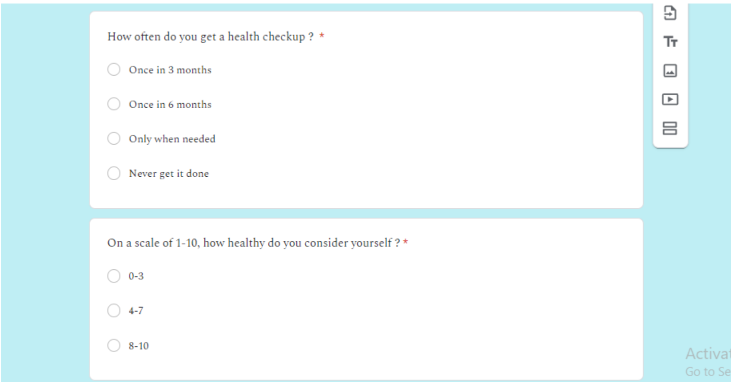


Age group *

- 15-20
- 21-30
- 31-40
- 41-60
- Above 60

Select your region of residence *

- Navi Mumbai
- Thane
- Mumbai
- Other

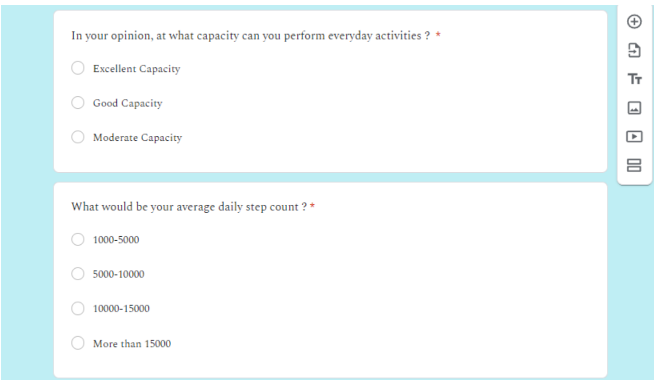


How often do you get a health checkup ? *

- Once in 3 months
- Once in 6 months
- Only when needed
- Never get it done

On a scale of 1-10, how healthy do you consider yourself ? *

- 0-3
- 4-7
- 8-10

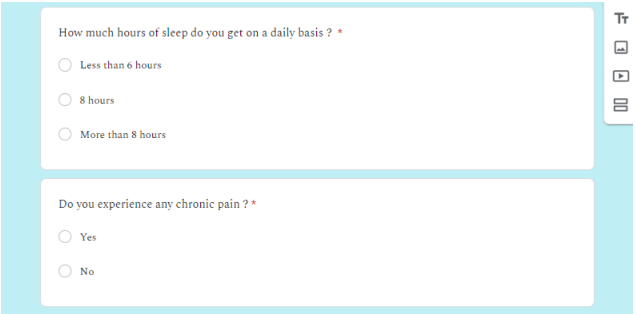


In your opinion, at what capacity can you perform everyday activities ? *

- Excellent Capacity
- Good Capacity
- Moderate Capacity

What would be your average daily step count ? *

- 1000-5000
- 5000-10000
- 10000-15000
- More than 15000

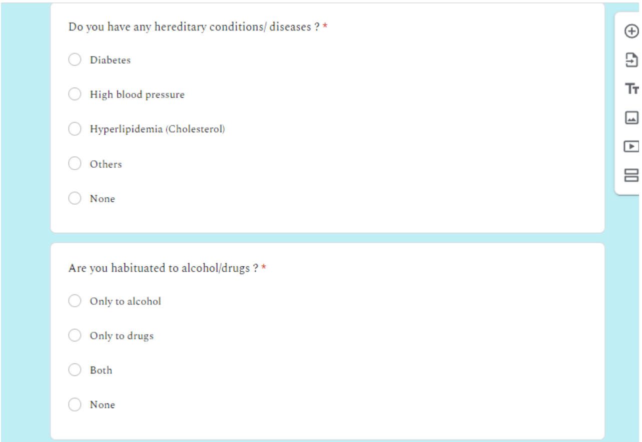


How much hours of sleep do you get on a daily basis ? *

- Less than 6 hours
- 8 hours
- More than 8 hours

Do you experience any chronic pain ? *

- Yes
- No

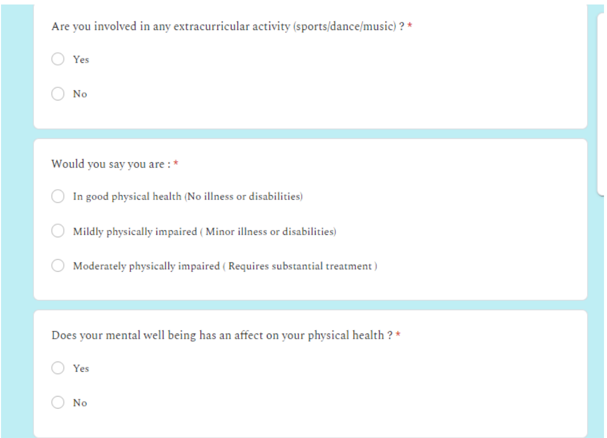


Do you have any hereditary conditions/ diseases ? *

- Diabetes
- High blood pressure
- Hyperlipidemia (Cholesterol)
- Others
- None

Are you habituated to alcohol/drugs ? *

- Only to alcohol
- Only to drugs
- Both
- None



Are you involved in any extracurricular activity (sports/dance/music) ? *

- Yes
- No

Would you say you are : *

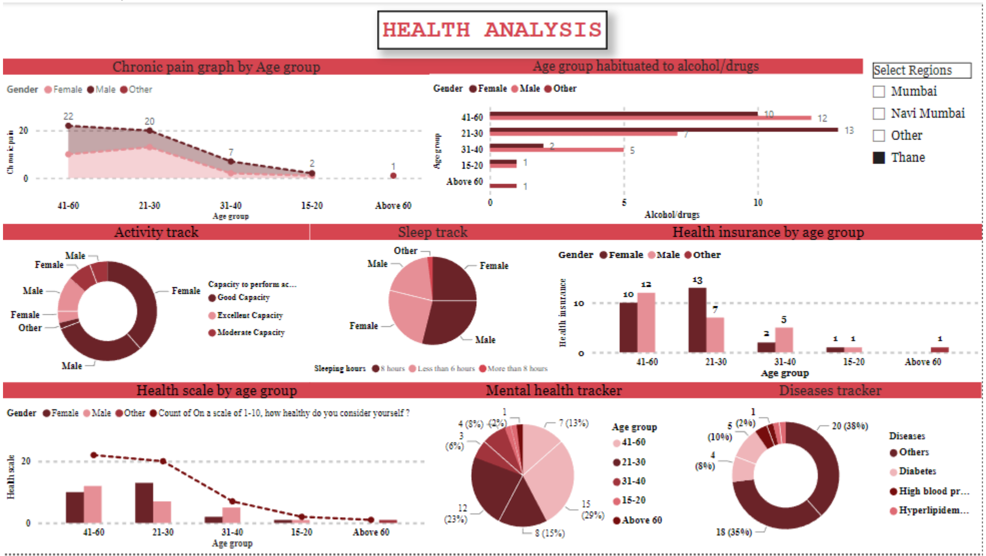
- In good physical health (No illness or disabilities)
- Mildly physically impaired (Minor illness or disabilities)
- Moderately physically impaired (Requires substantial treatment)

Does your mental well being has an affect on your physical health ? *

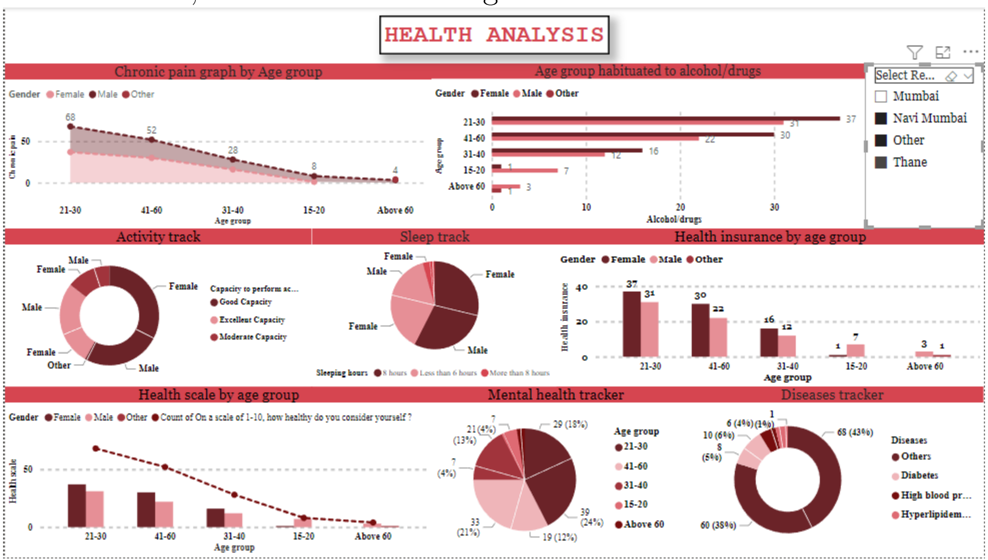
- Yes
- No

10.2 Region Wise

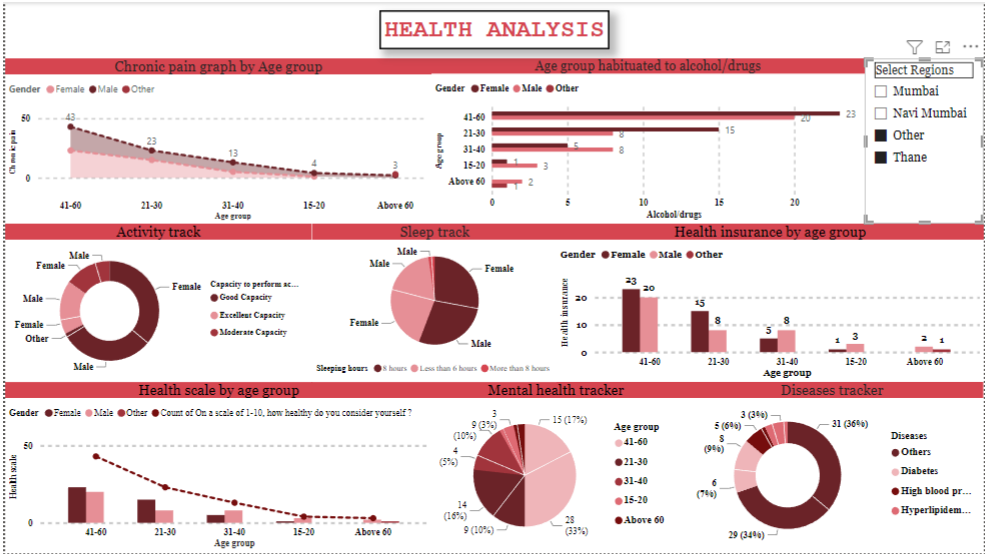
Thane Analysis:



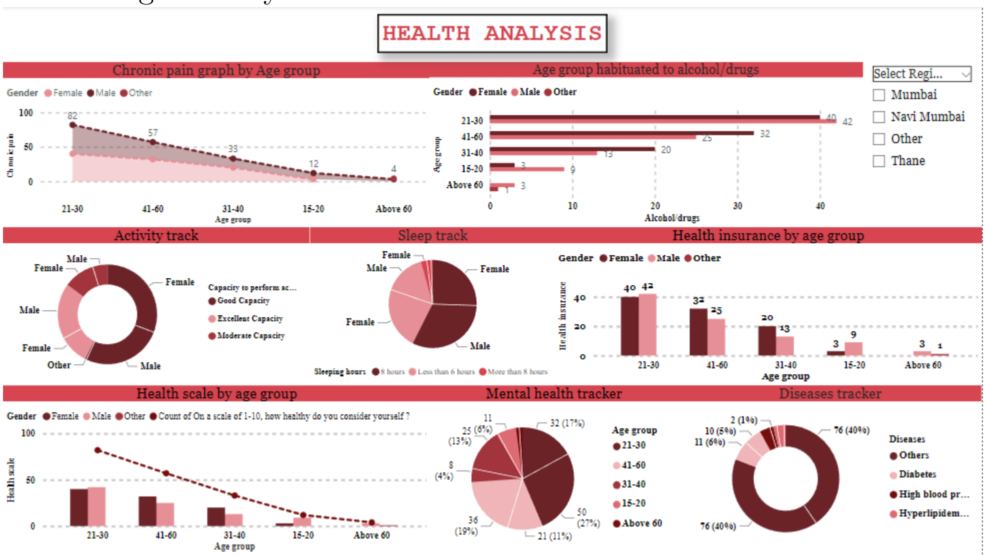
Navi Mumbai, Thane and Other regions:



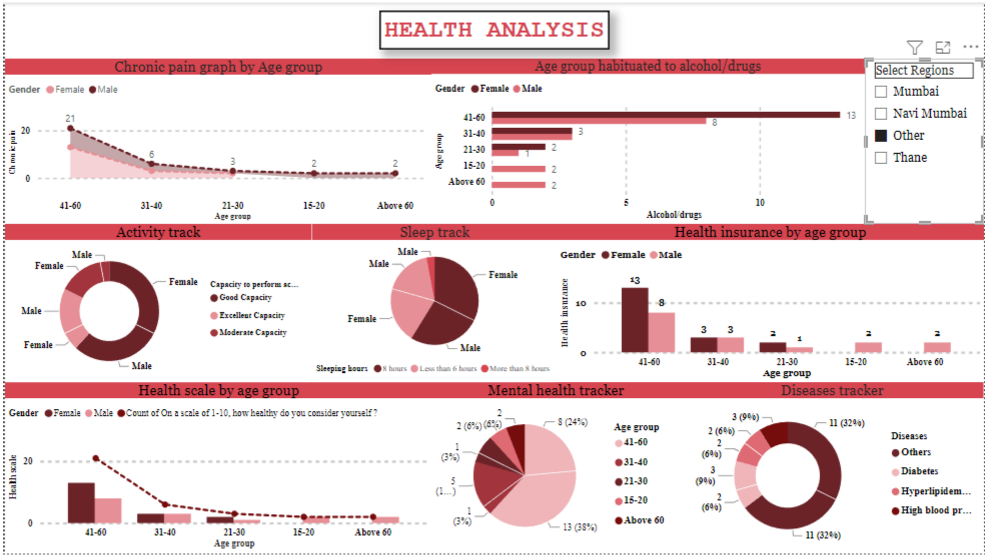
Thane and Other:



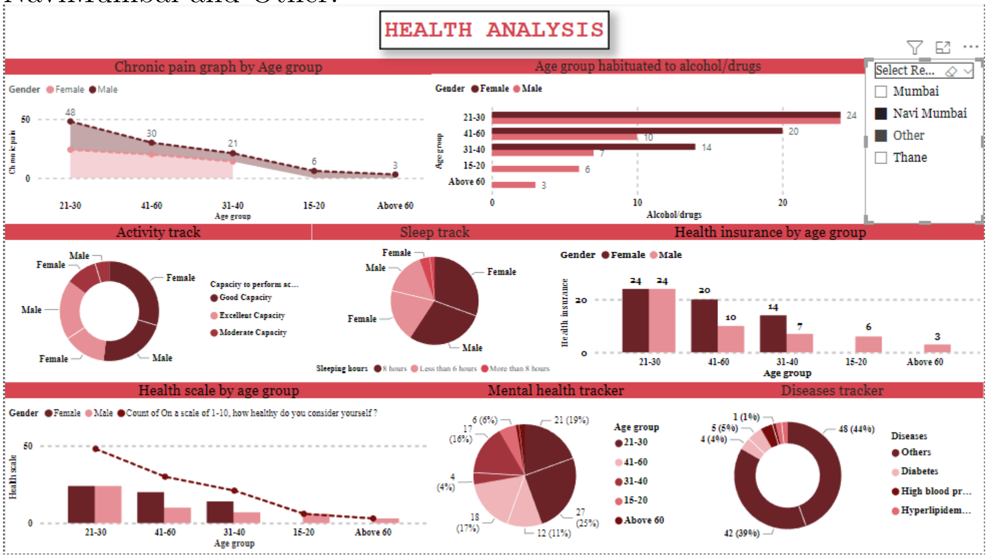
Overall region analysis:



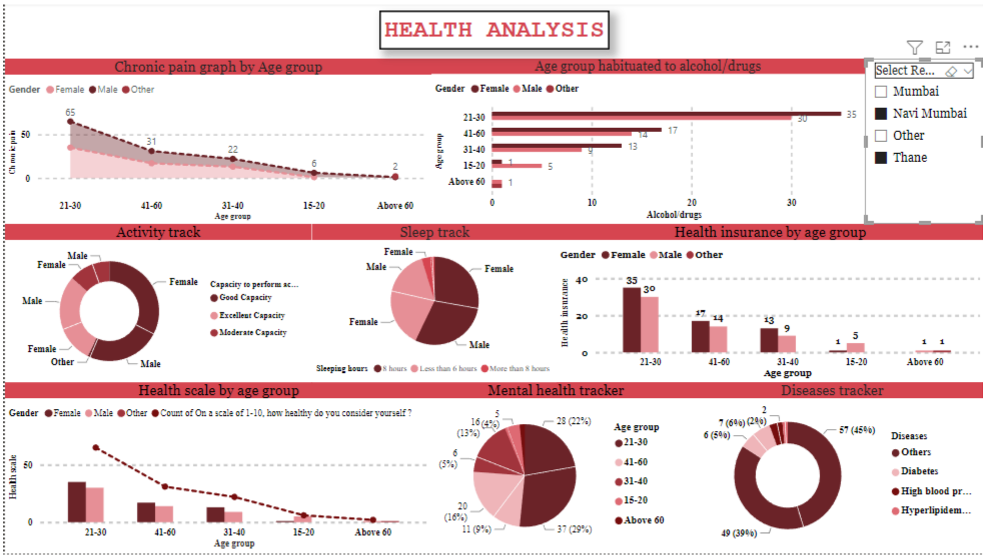
Other region analysis:



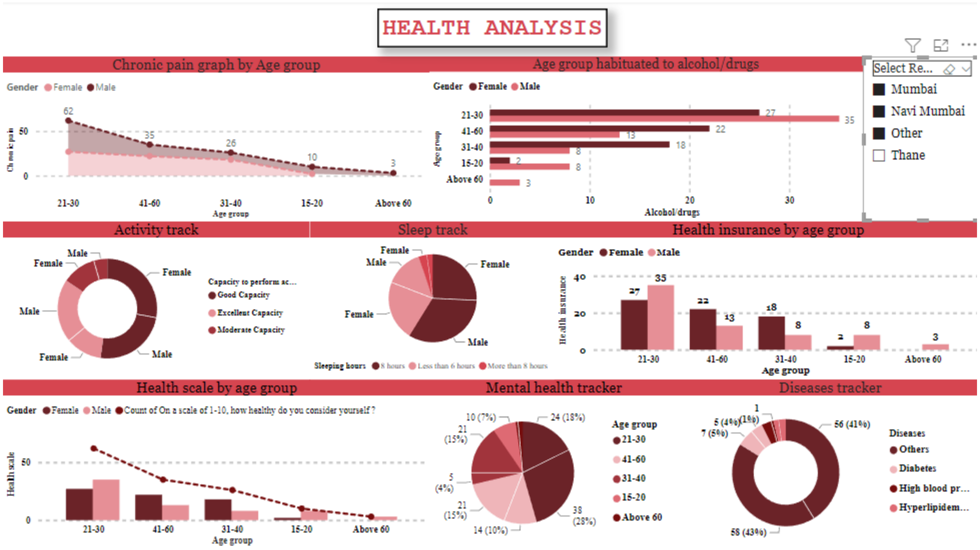
NaviMumbai and Other:



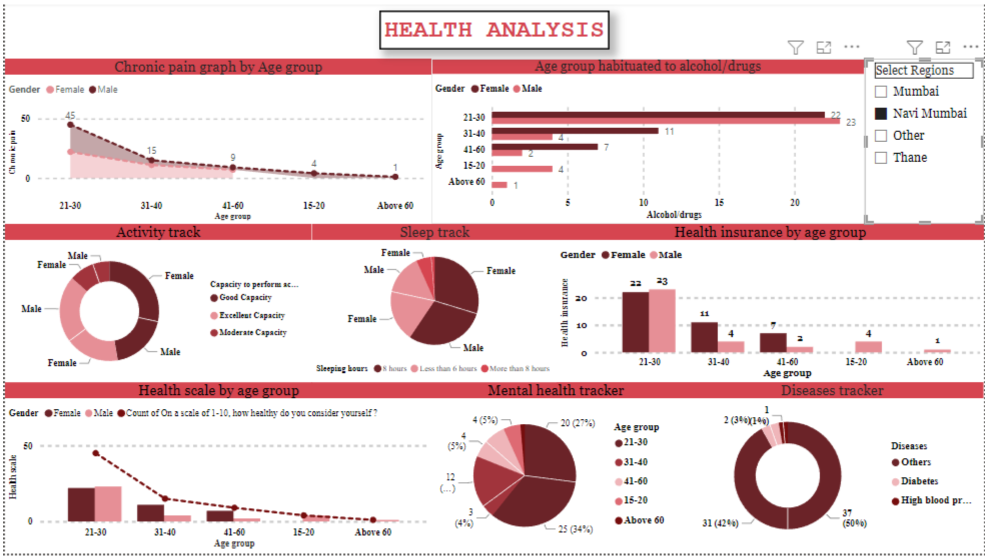
NaviMumbai and Thane :



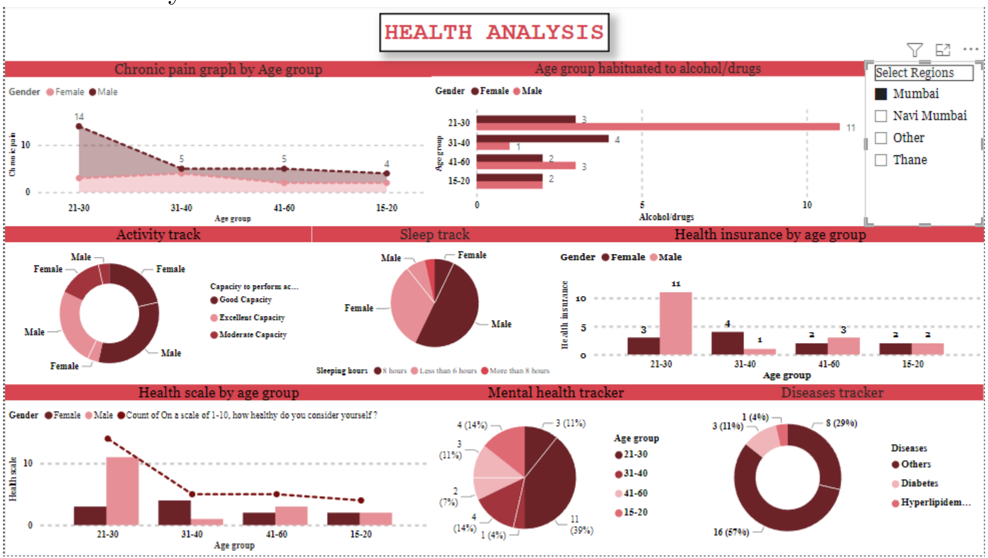
Mumbai,NaviMumbai and Other:



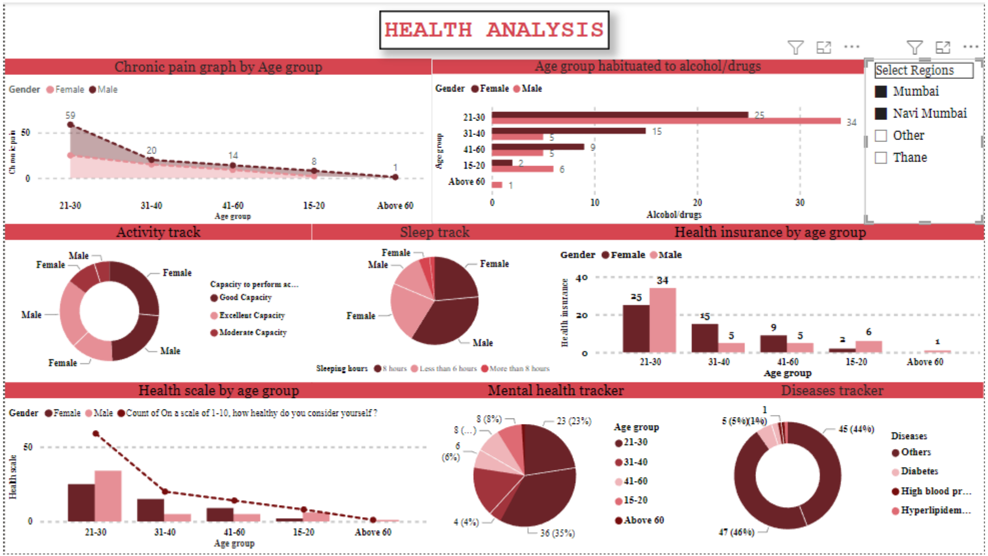
Navi mumbai analysis:



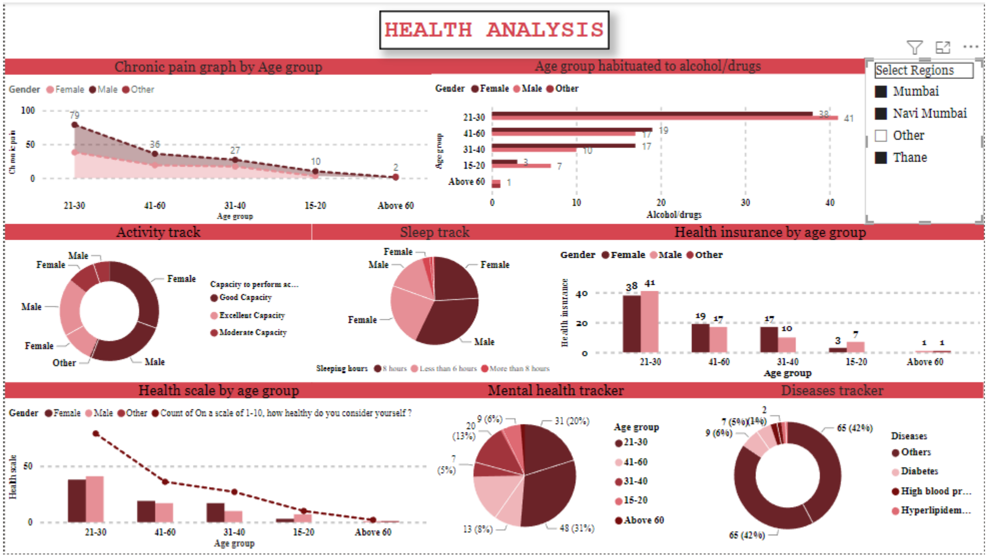
Mumbai anlysis:



Mumbai and Navi Mumbai :

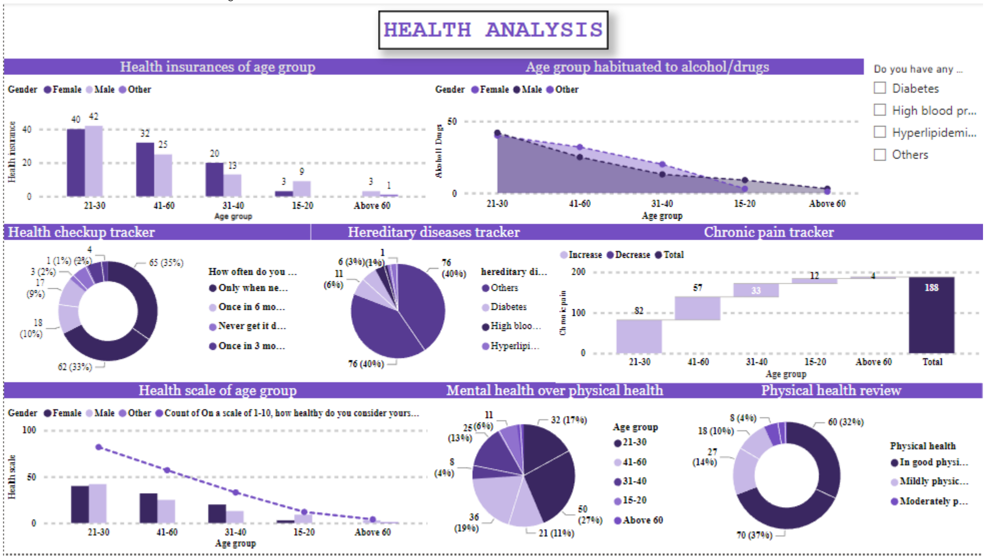


Mumbai, NaviMumbai and Thane :

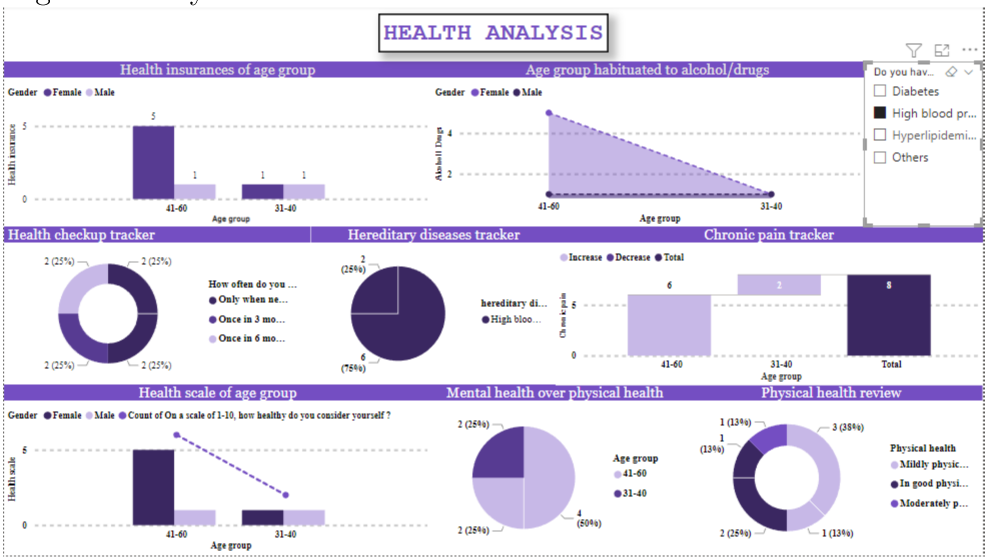


10.3 Disease Wise

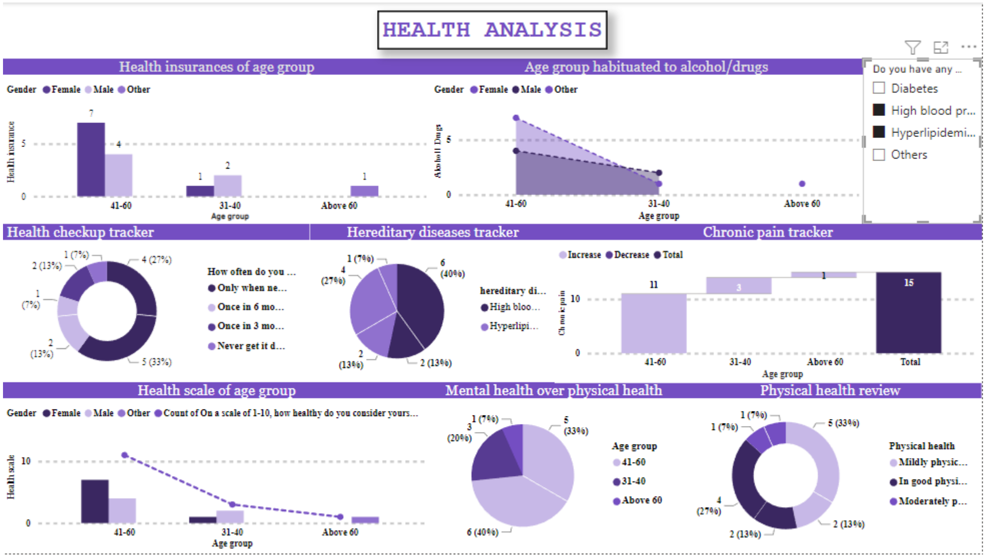
Overall disease analysis:



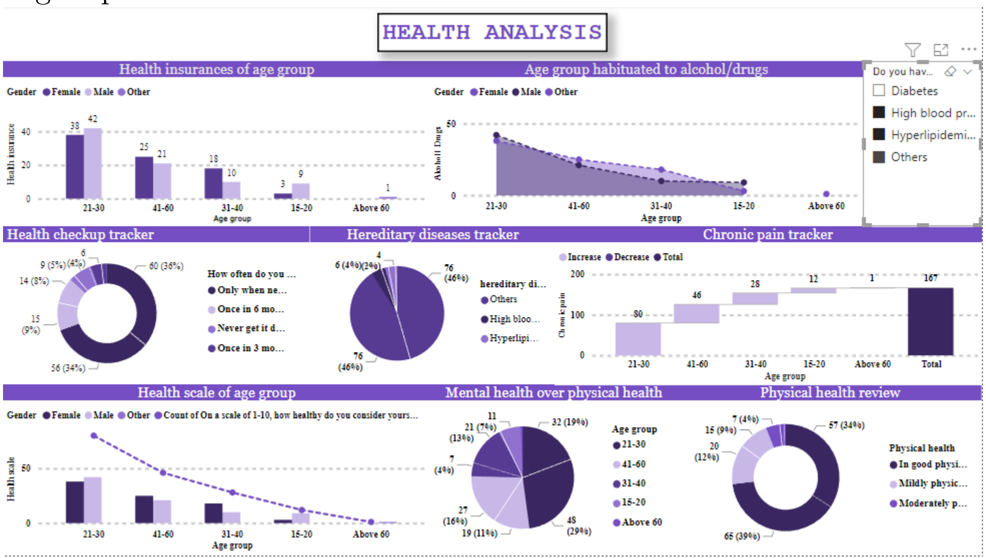
High BP analysis:



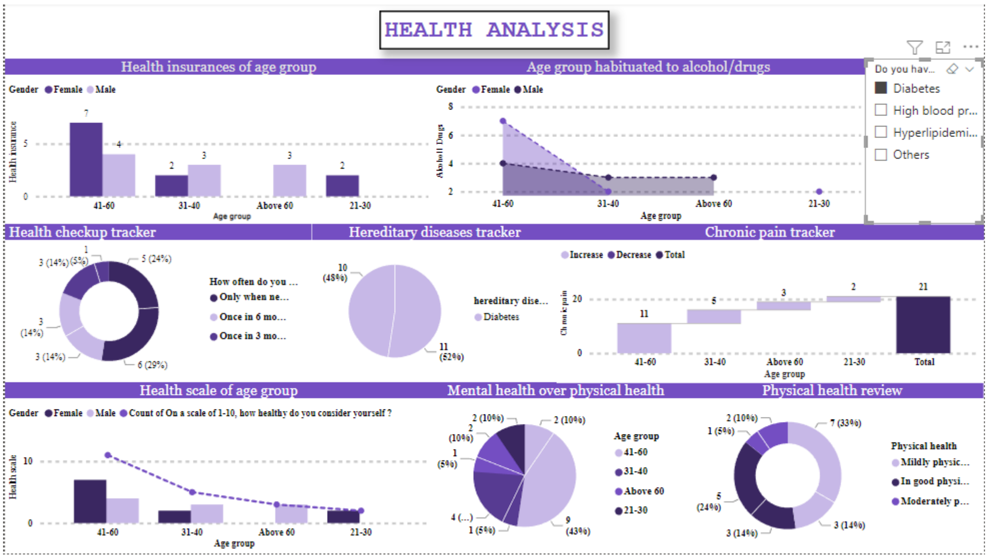
High Bp and Cholesterol



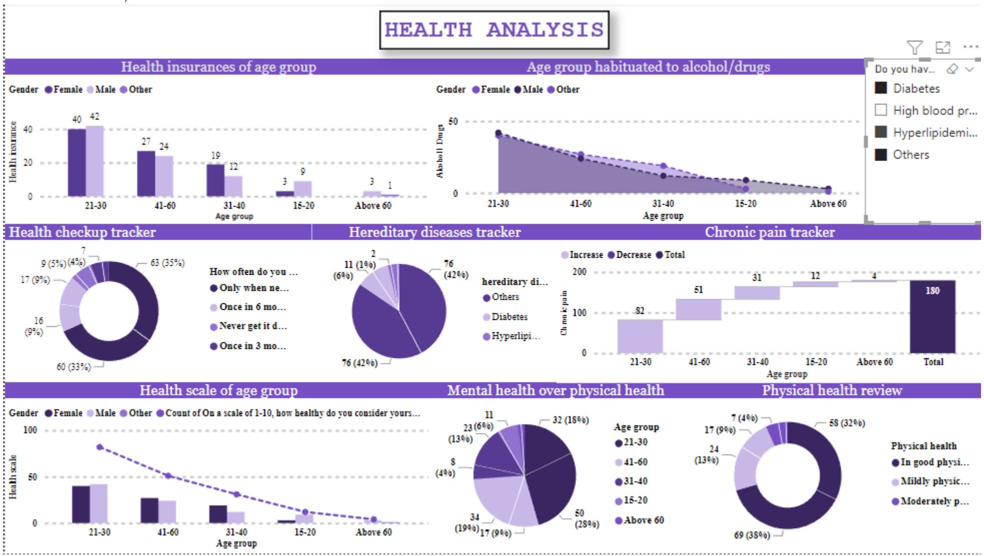
High Bp and Cholesterol and Others



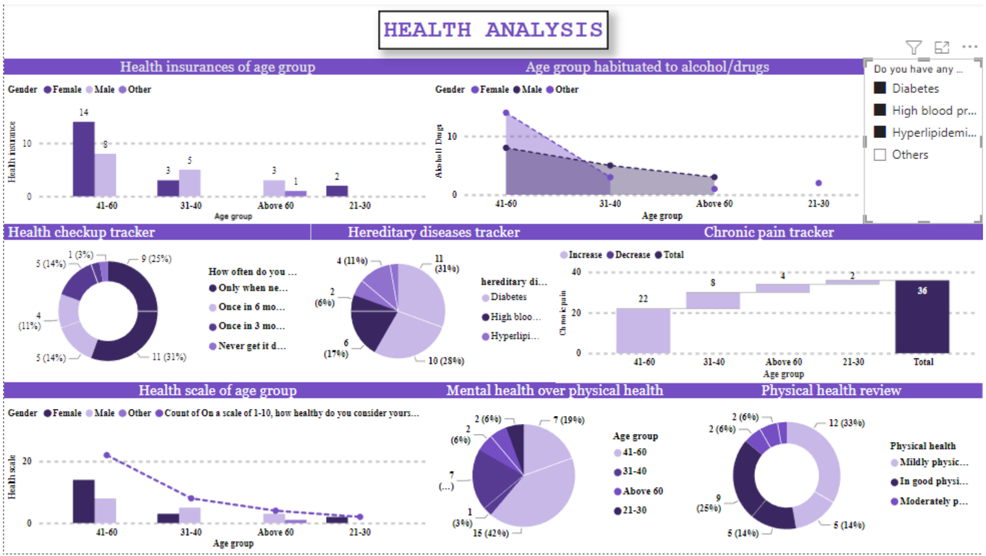
Diabetes Analysis:



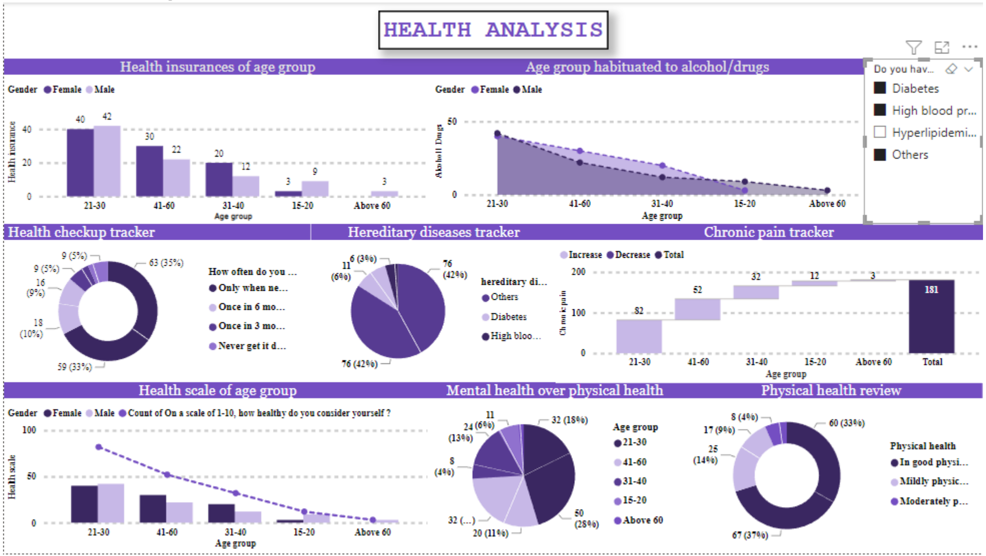
Diabetes, Cholesterol and Others



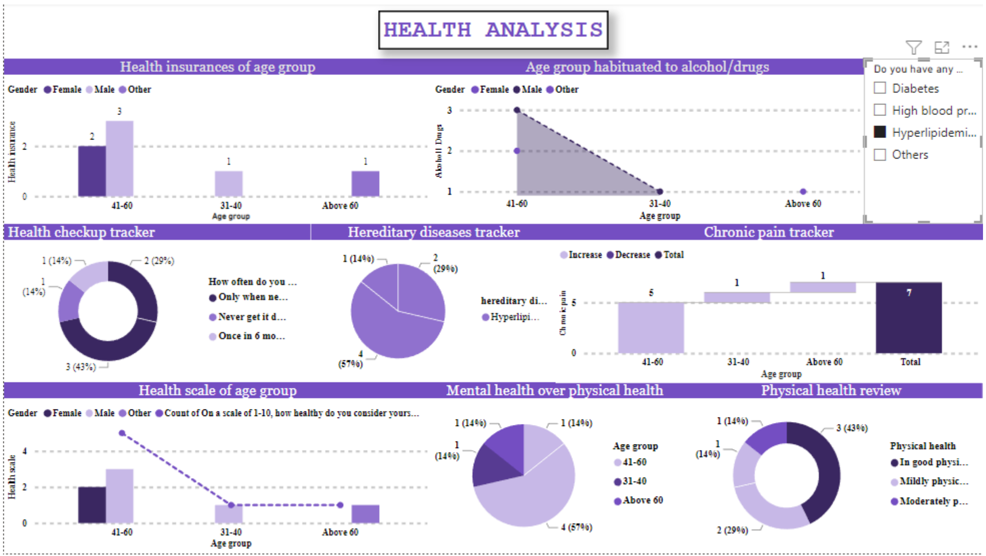
Diabetes, High BP and Cholesterol



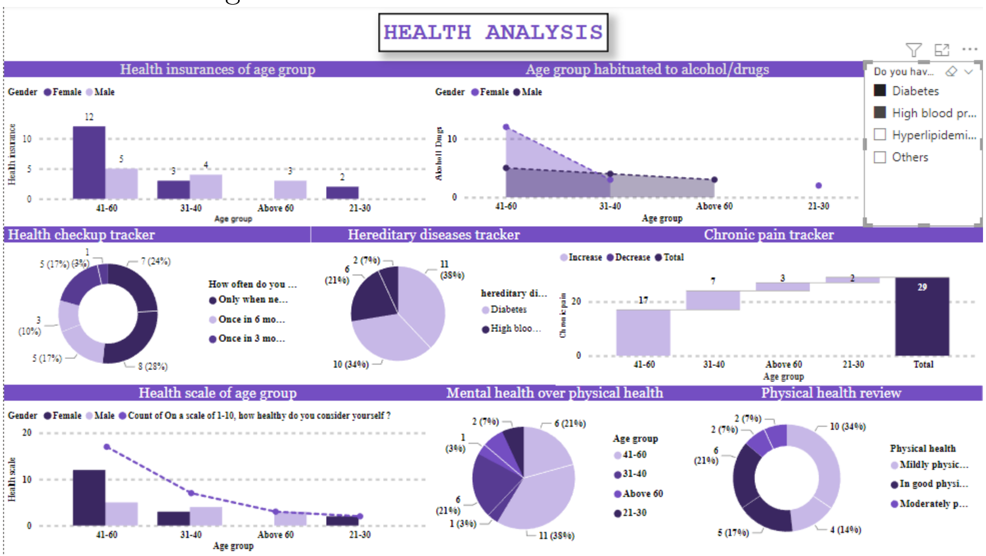
Diabetes,High Bp and Others



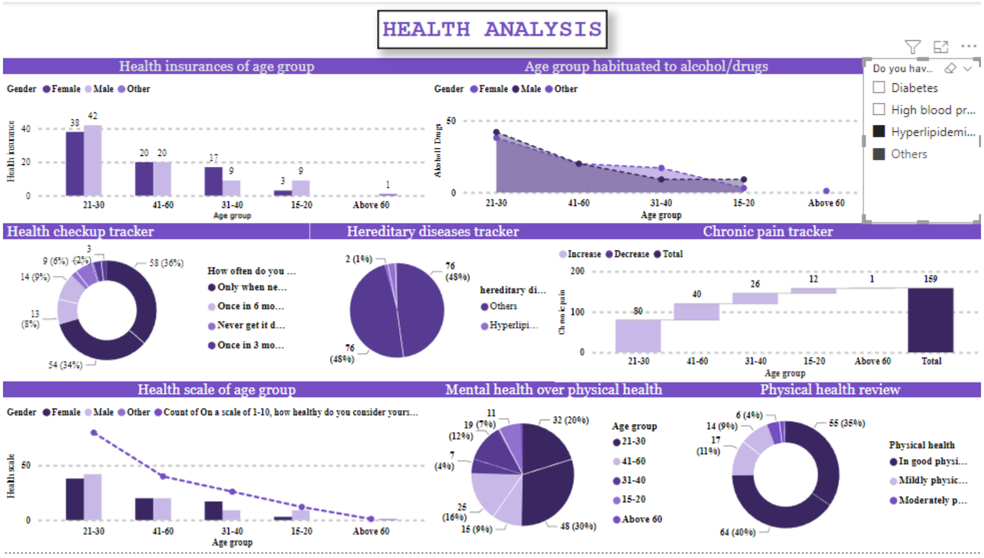
Cholesterol Analysis:



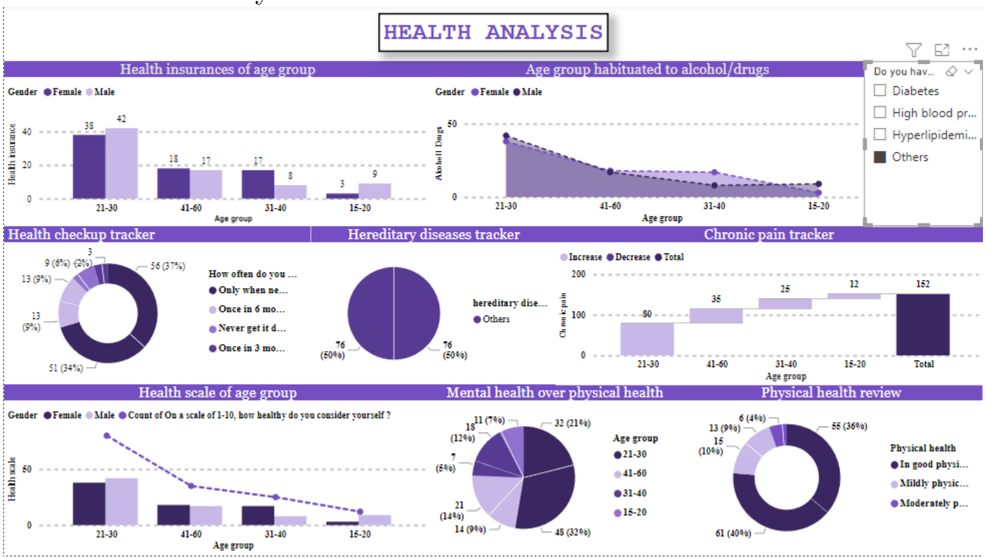
Diabetes and High BP:



Cholesterol and Others

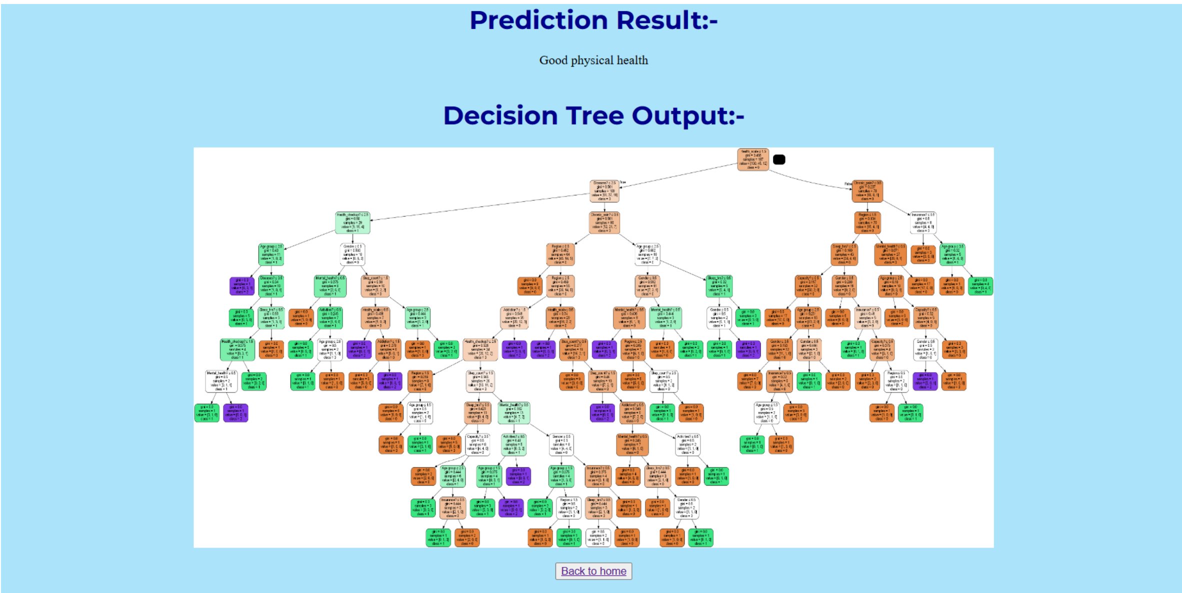


Other disease analysis:



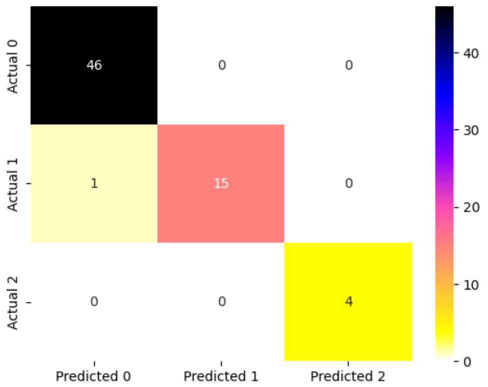
10.4 User Interface


How much hours of sleep do you get on a daily basis ?
Less than 6 hours
8 hours
More than 8 hours
Do you experience any chronic pain ?
Yes
No
Do you have any hereditary conditions/ diseases ?
Diabetes
High blood pressure
Hyperlipidemia (Cholesterol)
None
Are you habituated to alcohol/drugs ?
Only to alcohol
Only to drugs
None
Are you involved in any extracurricular activity (sports/dance/music) ?
Yes
No
Does your mental well being has an affect on your physical health ?
Yes
No
Do you have a health insurance ?
Yes
No
SUBMIT

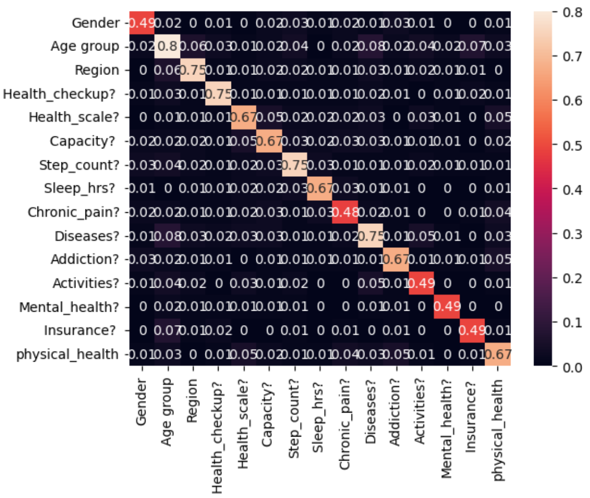

Prediction Result:-
Good physical health
Decision Tree Output:-
Back to home

10.5 Confusion Matrix

```
Out[49]: <AxesSubplot:>
```



10.6 Co-relation heat map



10.7 Classification report

```
In [50]: print(classification_report(y_test, y_pred))

              precision    recall  f1-score   support

           0       0.98      1.00      0.99        46
           1       1.00      0.94      0.97        16
           2       1.00      1.00      1.00         4

    accuracy                           0.98        66
   macro avg       0.99      0.98      0.99        66
weighted avg       0.99      0.98      0.98        66
```

Chapter 11

Conclusions

The project of gathering data of people from different age groups, regions, medical history and analyzing it has been completed successfully.
The region wise data can be visualized as well as disease wise data can be visualized using Microsoft Power Bi software.
On completing this project we can to know the basics of how to gather data from different sources and process it. The various pre-processing techniques were introduced to us.
The implementation of the CART algorithm was successful. The model can predict the physical health of an individual based on the attributes provided to the model.
On completing this project the concepts of machine learning and it's practical applications are more clear.

Chapter 12

Limitations

The limitations of the project is that on collecting the data of people, and analyzing it and creating a model based on the data, the model have not implemented concepts of artificial intelligence. The concept of ensemble learning can be implemented on the model.

Chapter 13

Future Scope of the Project

In future a model can be developed wherein, applications of artificial intelligence such as CNN or Ensemble learning can be implemented. A more accurate and efficient model can be developed with the use of artificial intelligence. We can apply different machine learning algorithms over the collected data and compare them to obtain the more efficient algorithm.

13.1 References

Websites referred:
https://www.google.co.in/
https://powerbi.microsoft.com
https://www.youtube.com/
https://www.wikipedia.org/